

Fig. 2: Actual-size PCB pattern of the low-noise, low-cost audio-distribution amplifier

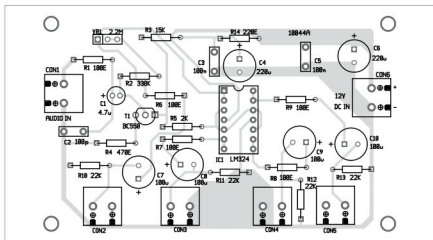


Fig. 3: Component layout of the PCB

feature of the circuit is the feedback implemented with R2 (330-kilo-ohm) and potentiometer VR1 (2.2-mega-ohm) between output of the first buffer (A1) and base of T1.

R2 and VR1 implement AC and DC negative feedback, which set the initial working point of T1 and reduce distortions in the output stages. Collector current of T1 is usually selected in the range 0.2mA to 1mA, depending on the power supply.

Voltage gain of the amplifier is set to around 30 and can be calculated with the approximate formula given below:

$$A_v = -R_3/R_4$$

Noise of the circuit depends practically only on the noise from

the preamplifier stage around low-noise T1. Consequently, you would have low noise for the circuit. Alternatively, you can use noisy, low-cost but high-output-current op-amps as output buffers.

Power supply. Circuit is operational in full operation range of the op-amp and transistor. For LM324, operating range is from 3V to 32V. Power supply above 9V is preferable. Quiescent current of the circuit is low and is equal to the emitter current of T1 plus quiescent current of the op-amp. 12V DC power supply was used during testing.

Construction and testing

An actual-size, single-side PCB pattern of the low-noise, low-cost

PARTS LIST

Semiconductors:

- IC1 - LM324 quad operational amplifiers
- T1 - BC550 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R6-R9 - 100-ohm
- R2 - 330-kilo-ohm
- R3 - 15-kilo-ohm
- R4 - 470-ohm
- R5 - 2-kilo-ohm
- R10-R13 - 22-kilo-ohm
- R14 - 220-ohm
- VR1 - 2.2-mega-ohm potentiometer

Capacitors:

- C1 - 4.7 μ F, 35V electrolytic
- C2 - 100pF ceramic disk
- C3, C5 - 100nF ceramic disk
- C4, C6 - 220pF, 35V electrolytic
- C7, C8, C9, C10 - 100pF, 35V electrolytic

Miscellaneous:

- CON1-CON6 - 2-pin terminal connector
- 12V DC power supply

audio-distribution amplifier is shown in Fig. 2 and its component layout in Fig. 3.

For testing, apply well-regulated 12V power supply. First, short the input at CON1 (with no input signal) and vary VR1 till you get around half the power supply (6V) at collector of T1. Then, open the shorting input and apply a sine-wave signal around 1kHz with 10mVpp amplitude at CON1.

You can obtain a sine-wave with peak-to-peak amplitude depending on the selected gain at output pins 1, 7, 8 and 14 of A1, A2, A3 and A4 op-amps, respectively. You should get around 300mVpp in all output pins.

$$V_{out} = (-R_3/R_4) \times V_{in} \\ = 30 \times 10\text{mVpp} = 300\text{mVpp}$$

You can replace VR1 and R2 with a single resistor of around 510-kilo-ohm if the power supply is 15V and transistor T1 is BC550C. **EFY**



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